

AVI GARG

Senior Software Engineer (C++ | Unreal Engine | Gameplay | Graphics)

📞 +91 93540-89308 @ gargav@tcd.ie 📄 Portfolio: pixelsenseiavi.github.io 📄 github.com/PixelSenseiAvi 📍 Jaipur, IN

SUMMARY

Gameplay Engineer with 5+ years building games and interactive applications in Unreal Engine and Unity, currently owning combat and vehicle systems on two UE5 titles. Architected a Gameplay Ability System combat framework spanning 32 abilities, combo chains and 14 enemy AI archetypes, plus a Chaos Vehicle system with tunable handling, destruction and mounted weapons, replicated for multiplayer at a locked 60 FPS. Builds data-driven systems designers can tune without engineering support, and profiles relentlessly with Unreal Insights to protect frame budgets. Previously core programmer on three Unity architectural visualization apps. Strong C++ and C# foundation with deep roots in real-time rendering and OpenGL, and an MSc in Computer Graphics and Vision from Trinity College Dublin.

EXPERIENCE

Game Developer

Morphionix (Contract)

📅 10/2025 - Present 📍 Remote

Gameplay Engineer on TankMates, a physics-based co-op platformer, and an untitled third-person action-combat title. Owned core tank and combat gameplay systems, taking TankMates from prototype to release, working alongside designers, animators and QA to deliver responsive, data-driven mechanics that held 60 FPS on PC.

Combat & Ability Systems (Untitled Action Title)

- Designed and implemented a modular melee and ranged combat framework in C++ using the Gameplay Ability System (GAS), covering 32 player abilities, combo chains, hit reactions, stagger, and status effects.
- Built a data-driven damage and hitbox pipeline (Animation Notifies, Gameplay Tags, Gameplay Effects) so designers could tune weapons and enemies in Data Assets without engineering support, cutting iteration time by roughly 40%.
- Authored enemy combat AI with Behavior Trees, State Trees, and the Environment Query System (EQS), including target selection, flanking, and cooldown-based attack scheduling for 14 enemy archetypes.
- Implemented lock-on targeting, dodge and parry windows, and camera-shake/hitstop feedback tuned through playtests to reach an 87% positive combat-feel rating in internal surveys.
- Prototyped and balanced 6 boss encounters with multi-phase state machines, telegraphed attacks, and difficulty scaling.

Tank Physics & Co-op Systems (TankMates)

- Built the physics-driven tank controller in C++ on Chaos Physics, tuned for precision platforming across chaotic, military-themed obstacle courses.
- Engineered the shared two-player control scheme, splitting driving and turret aiming/firing between players, with a single-player mode covering both roles.
- Implemented online co-op with replicated tank physics, turret and firing state, plus local co-op, full controller support, and input remapping through Enhanced Input.
- Developed enemy tank and turret AI using Behavior Trees and AI Perception, with targeting, firing patterns, and line-of-sight checks.
- Built destructible obstacles and data-driven hazard, checkpoint, and respawn systems exposed through Data Assets, so designers could build and tune levels without engineering support.
- Tuned Lumen lighting, shadows, and post-processing, profiling with Unreal Insights to hit visual targets while holding 60 FPS on PC.

Cross-Team Impact

- Shipped 9 milestones on schedule; served as point of contact for combat and vehicle features across design, animation, audio, and QA.
- Wrote technical design docs and tuning guides adopted as the team standard, and mentored 2 junior engineers on the Gameplay Ability System and vehicle physics.
- Maintained CI-friendly gameplay tests (Unreal Automation Framework and Functional Tests) that caught 40+ regressions.

Tech: Unreal Engine 5, C++, Blueprints, Gameplay Ability System, Chaos Vehicles & Chaos Physics, Behavior Trees / State Tree / EQS, Animation Blueprints, Niagara, Enhanced Input, Network Replication, Unreal Insights & Unreal Automation Framework, Perforce/Git

Senior Unity Engineer

A2.AI Pvt. Ltd.

📅 04/2024 - 10/2025 📍 Adelaide, AU (Remote)

Built an iOS fitness app in Unity and C# that tracks workouts in real time using on-device AI. Integrated a MediaPipe pose-estimation pipeline for live skeletal tracking on mobile hardware, then built rep counting on top of it using joint-angle analysis and a state machine to detect exercise phases. Added a posture-feedback system that flags form deviations mid-rep, and optimized the ML inference pipeline to sustain 30+ FPS under mobile performance constraints.

Tech: Unity, C#, iOS (Xcode), MediaPipe, Pose Estimation

Senior Software Engineer (C++)

FEV Group

📅 05/2022 - 05/2024 📍 Pune, India

Senior C++ Developer working across multiple embedded and simulation teams to deliver real-time automotive software, spanning QNX instrument-cluster rendering, DDS middleware, Unreal Engine-based ADAS validation, and CI/CD automation.

- Designed and developed a real-time automotive instrument cluster - tachometer, speedometer, and full driver dashboard; using OpenGL SC 2.0 with custom GLSL shaders on QNX embedded hardware, achieving a 40% improvement in rendering and memory performance through GPU profiling and draw-call optimization.
- Spearheaded the team-wide transition from OpenDDS to CycloneDDS, leading end-to-end planning, compatibility testing, and deployment, improving runtime scalability and communication efficiency by 15% across the automotive real-time stack.
- Integrated CARLA simulator with the DDS middleware stack inside Unreal Engine, enabling real-time sensor data streaming into ADAS pipelines and closed-loop ACC validation across varied traffic scenarios without requiring physical vehicle hardware.

Tech: C++, OpenGL SC / GLSL, QNX, CycloneDDS, Unreal Engine, CARLA, ADAS, TCP/UDP, Azure CI/CD

R&D Immersive Developer

Modelworks.Studio

📅 10/2020 - 09/2021 📍 Dublin, IE (On-site)

Unity and Unreal Engine Developer building immersive real-estate and interactive experiences across multiple client projects, spanning VR property tours, city-scale geospatial visualization, and projection-mapped physical installations.

- Shipped 3 immersive real-estate applications (Apartment Finder, Leasing App, and VR Property Tour), enabling clients to explore listings and close leases, all built in Unity.
- Engineered real-time 3D geospatial visualization pipelines in Unreal Engine, integrating Cesium and OpenStreetMap to render city-scale and site-specific environments.
- Designed projection-mapped installations with custom Arduino hardware integrations for showrooms, events, and interactive experiences.
- Architected scalable systems that expanded client project scope by 20%, adding support for multiple properties and new platforms.

Tech: Unity, Unreal Engine, Three.js

Realtime Filmmaking Intern

Viga Entertainment Technology

📅 03/2020 - 05/2020 📍 Remote

Developed a blendshape animation tool using direct vertex manipulation, enhancing realism and production efficiency by over 30% and improving facial realism in virtual production pipelines.

Tech: C++, OpenGL, ImGui

R&D Digital Image Processing Intern

Defence Laboratory, DRDO

📅 12/2017 - 06/2018 📍 Jodhpur, India

Researched image-processing applications spanning object detection, pattern matching, segmentation, and machine learning. Built OpenCV preprocessing pipelines for thresholding, morphological operations, and feature extraction.

Tech: Python, OpenCV

EDUCATION

MASTERS IN SCIENCE in Computer Graphics and Vision (MSc)

Trinity College Dublin

📅 09/2018 - 08/2019 📍 Dublin, Ireland

- Coursework: Realtime Rendering, Realtime Animation, Computer Graphics, AR/VR, Advanced Software Engineering, Machine Learning, Computer Vision, Data Visualization, Mathematics of Light & Sound
- Thesis title: Evaluation of knot-tying techniques in medical settings using Deep Learning techniques

BACHELOR OF TECHNOLOGY in Computer Science

JECRC University

📅 08/2014 - 07/2018 📍 Jaipur, India

- Coursework: Programming in C, OOP (C++), Engineering Mathematics, Engineering Physics, Data structure and Algorithm, Discrete Maths, DBMS